

Scale-invariant satellization: a power-law model of node dominance verified across 502 cases in 11 domains

Short title: Scale-invariant satellization across 11 domains

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ABSTRACT

We propose Shadow Node Theory (SNT), a formal model of node satellization in which the ratio of productive output between a dominant hub and a peripheral node follows a power law $R(t) = a \cdot t^b$. The exponent b measures satellization velocity; $b > 0$ indicates active satellization; $b < 0$ indicates convergence or leapfrog. We verify the model across a corpus of 502 empirically confirmed cases in 11 domains — historical cities ($n=64$), country pairs ($n=230$), intra-national regions ($n=64$), digital ecosystems ($n=53$), biological invasion ($n=20$), predator-prey ($n=4$), parasite-host ($n=20$), planetary systems ($n=14$), stellar binaries ($n=8$), black hole accretion ($n=13$), and galactic systems ($n=12$) — spanning 30 orders of magnitude of temporal scale.

Three findings emerge from statistical analysis of the corpus. First, abrupt triggers produce significantly faster satellization than gradual ones (Mann-Whitney $U = 24,802$, $p = 1.91 \times 10^{-5}$, $n = 486$), a result stable across three successive corpus expansions. Second, institutional friction is the

dominant predictor of the exponent b across all substrates: domains without friction (biological invasion $b = +1.44$; parasite-host $b = +1.15$; planetary formation $b = +1.03$) produce $b > 1$, while high-friction domains (sovereign countries $b = -0.10$; predator-prey $b = +0.10$) produce $b \approx 0$. Third, political sovereignty and ecological mutual interdependence are equivalent satellization brakes, both preventing definitive satellization by making complete node extinction destructive to the hub.

We also present: (i) an N-body matrix correction for the Mexican national system (32 entities, INEGI 2022) showing the binary model underestimated Tlaxcala's satellization gradient by 9.3 $\times$; and (ii) the Atomic Sovereignty Index (ASI), operationalized on behavioral data from 4,774 users (precision = 1.0, zero false positives). The model is accompanied by eight falsifiability criteria and a four-step diagnostic protocol.

INTRODUCTION

Why do some regions remain poor despite decades of policy intervention? Why do some nations converge toward global leaders while others diverge irreversibly? Why does a black hole accreting in quasar mode follow a mathematically similar trajectory to the Norway rat displacing the black rat from Europe? These questions share a structural feature that aggregate models consistently fail to capture: the dynamics of resource extraction between proximate nodes operating at different hierarchical levels.

Standard development economics models the poverty trap as a problem of insufficient capital accumulation [1]. Institutional models attribute persistent inequality to extractive institutions [2]. Both approaches capture part of the phenomenon but do not formalize the rate of divergence between specific nodes within a system. We propose a different mechanism: satellization — the progressive extraction of productive residual energy from a peripheral node by a dominant hub — whose dynamics follow a power law invariant to temporal scale and substrate.

The preferential attachment mechanism [3] establishes that scale-free networks emerge when new connections form with probability proportional to existing degree, generating the 'rich get richer' dynamic. Leapfrogging theory [4] identifies conditions under which peripheral nodes can bypass dominant ones through orthogonal-dimension investment. The Free Energy Principle [5] formalizes how biological systems minimize surprise. Complex Adaptive Systems theory [6] describes emergent behavior from local rules. The present work bridges these literatures by formalizing satellization as a quantifiable power-law process and verifying it across domains ranging from historical city-pairs to galactic mergers.

Three gaps motivate this work. First, no existing model quantifies satellization dynamics across historical cities, nation-states, intra-national regions, digital platforms, ecological communities, and astrophysical systems within a unified formal framework. Second, trigger taxonomy in leapfrogging theory does not formalize the velocity difference between abrupt, hybrid, and gradual events. Third, no operational index of cognitive sovereignty exists for the individual node computable from observable behavioral data without self-report. This paper addresses all three gaps.

MATERIALS AND METHODS

Formal model

A Shadow Node (peripheral node) is any system component whose productive output is systematically extracted by a dominant hub over time. The satellization ratio $R(t) = \text{production_hub}(t) / \text{production_shadow}(t)$ is the central observable. When $R(t)$ follows a power law $R(t) = a \cdot t^b$ with $b > 0$, satellization is active. When $b < 0$, convergence or leapfrog is occurring. The exponent b classifies dynamics: $b > 0.45$ (accelerated satellization, abrupt trigger class); $0.1 < b < 0.45$ (gradual satellization); $-0.1 < b < 0.1$ (approximate steady state or mutual interdependence); $b < -0.1$ (convergence or successful leapfrog).

The institutional friction hypothesis predicts that the primary determinant of b is not substrate or temporal scale but the degree to which resource transfer is mediated by regulatory, political, or ecological constraints. In the absence of friction, dominant nodes extract from peripheral ones at rates determined only by their mass advantage, producing $b > 1$. In the presence of friction, complete satellization would destroy the hub, and the system equilibrates at $b \approx 0$. This hypothesis generates a directly testable prediction: domains without institutional mediation should produce systematically higher b values than domains with mediation.

Case selection criteria

Cases were included if they satisfied four conditions: (a) two nodes in critical proximity within the same system; (b) an identifiable trigger event or process; (c) production or population data at a minimum of four temporal points; and (d) the satellization process has either completed or is actively ongoing. No cases were excluded based on outcome direction — both satellization ($b > 0$) and convergence/leapfrog ($b < 0$) cases are included to avoid confirmation bias.

Analytical pipeline

For each case pair: (1) construct the satellization ratio time series $R(t)$; (2) apply log-log linearization: $\log(R) = \log(a) + b \cdot \log(t)$; (3) estimate a and b by ordinary least-squares; (4) compute R^2 and Pearson correlation in log space for significance testing; (5) classify by SNT taxonomy. For the N-body Mexican matrix, we additionally compute the composite gradient — the total satellization force on each node accounting for all higher-level hubs, not only the immediate one. All code is available in the replication package [7].

Data sources

Domain A (historical cities, $n = 64$): Bairoch, Batou & Chèvre [8], 2,200 European cities 800–1850. Domain B (country pairs, $n = 230$): Maddison Project Database 2023 [9], GDP per capita 1–2018 AD. Domain C (intra-national regions, $n = 64$): OECD Regional Database, INEGI SCNM

[10], Eurostat NUTS2, U.S. BEA state GDP. Domain D (digital ecosystems, $n = 53$): StatCounter, Statista, IDC, SEC EDGAR, HackerEarth 2026 (proprietary dataset; aggregate results available in replication package [7]). Domains E1–E3 (biological, $n = 44$): species population databases, WHO AMR surveillance, UNAIDS longitudinal data, Hudson Bay Company fur records [11–15]. Domains F1–F4 (astronomical, $n = 47$): NASA Exoplanet Archive, Gillessen et al. [16], Erkal et al. [17], Majewski et al. [18], Pollack et al. [19].

Statistical tests

H1 (power-law universality): verified for each case individually via $R^2 > 0.7$ criterion. H2 (trigger-type effect on velocity): Mann-Whitney U test (one-tailed, abrupt > gradual), excluding mutual interdependence domains E2 and F4. H3 (institutional friction effect): Kruskal-Wallis test across all domains, then restricted to friction categories. H4 (domain invariance): Kruskal-Wallis across all 11 domains. Significance threshold: $\alpha = 0.05$. All tests implemented in Python 3.11 using `scipy.stats` [7].

Falsifiability criteria

We specify eight criteria under which each postulate would be falsified. RC1 (scalar velocity): falsified if a technology is systematically adopted faster by firms than by individuals. RC2 (immune response): falsified if hub adaptation toward peripheral nodes is documented more frequently than suppression. RC3 (qualitative inextractability): falsified if hubs systematically neutralize node knowledge differentials. RC4 (dual minimum threshold): falsified if leapfrog sustains itself with either quantitative or qualitative resources below operational minimum. RC5 (expansion sequence): falsified if direct expropriation produces more stable outcomes than silent absorption. RC6 (irreversibility): falsified if a Shadow Node reverses satellization endogenously without exogenous trigger. RC7 (ASI operationalization): falsified if users with $ASI > 1$ show outcomes comparable to users with $ASI < 0.5$ in cognitive sovereignty tasks. RC8 (mutual

interdependence brake): falsified if predator-prey or sovereign-state systems produce sustained $b > 0.5$ without exogenous perturbation.

RESULTS

Corpus overview

Of 502 cases, 156 (31.1%) are statistically significant ($p < 0.05$). Cases with $b > 0$ (active satellization): 237. Cases with $b < 0$ (convergence or leapfrog): 262. Range of b : $[-2.852, +7.086]$ across 30 orders of magnitude of temporal scale. The 31.1% significance rate reflects the inclusion of domains with structural $b \approx 0$ (sovereign countries, predator-prey, stellar binaries in equilibrium) where the model correctly predicts near-zero velocity rather than divergence.

The distribution of mean exponent b by domain is presented in Fig 1. The ordering of domains by b_{mean} corresponds precisely to their level of institutional friction (Kruskal-Wallis $H = 38.9$, $p < 0.0001$ across 11 domains): biological invasion ($b = +1.44$), parasite-host ($b = +1.15$), planetary formation ($b = +1.03$), black holes ($b = +0.65$), galactic systems ($b = +0.30$), digital ecosystems ($b = +0.30$), stellar binaries ($b = +0.11$), predator-prey ($b = +0.10$), intra-national regions ($b = +0.05$), historical cities ($b = +0.07$), and sovereign country pairs ($b = -0.10$). This ordering is the central empirical finding of the extended corpus.

Fig 1. Power-law satellization dynamics across 502 verified cases in 11 domains. (A)

Distribution of exponent b by domain, ordered by mean b . (B) Percentage of cases with statistically significant power-law fits ($p < 0.05$) per domain. (C) Mean exponent b by institutional friction category. All colors are consistent across panels. Error bars: ± 1 SE.

Finding 1: abrupt triggers produce faster satellization

Mann-Whitney test on 486 valid cases (excluding mutual interdependence domains E2 and F4): $U = 24,802$, $p = 1.91 \times 10^{-5}$. Mean exponents: abrupt $b = +0.552$ ($n = 128$), hybrid $b = +0.200$ ($n = 48$), gradual $b = -0.013$ ($n = 310$). The trigger-type hierarchy (abrupt $>$ hybrid $>$ gradual) has remained stable across three successive corpus expansions from 57 to 114 to 502 cases. An earlier finding from the 57-case corpus suggested hybrid triggers were fastest ($p = 0.0098$); the 502-case result definitively corrects that artifact of small N . The distribution of b by trigger type is shown in Fig 2B.

Fig 2. Institutional friction as primary predictor of satellization velocity. (A) Mean exponent b against institutional friction rank for all 11 domains. Point size proportional to n . Dashed line: OLS regression. (B) Distribution of b by trigger type (abrupt, hybrid, gradual), excluding domains E2 and F4. Statistical test result annotated. (C) Normalized domain metrics (b_{mean} , % significant, $\log n$) as a heatmap ordered by mean b .

Finding 2: institutional friction predicts b

Domains without institutional friction (E1, E3, F1) produce $b_{\text{mean}} > 1.0$ in all cases. Domains with high institutional friction (B, E2, F2) produce $b_{\text{mean}} \approx 0$. The astrophysical corpus provides a particularly clean test: among black hole systems (Domain F3), accretion mode — not black hole mass — predicts b . M87* (mass: $6.5 \times 10^9 M_{\odot}$, radiatively inefficient accretion flow mode): $b \approx 0$. TON 618 (mass: $6.6 \times 10^{10} M_{\odot}$, quasar mode): $b = +6.498$. M87* is ten times less massive than TON 618 yet produces b two orders of magnitude lower. RIAF accretion introduces self-limiting friction into the mass transfer process; quasar-mode accretion operates without such friction. This constitutes an astrophysical analogue of institutional friction.

Finding 3: political sovereignty equals ecological interdependence as braking mechanism

With 230 sovereign country pairs (Domain B), $b_{\text{mean}} = -0.098$. Country pairs with GDP per capita ratios of $7\times$ to $100\times$ produce $b \approx 0$ when both nodes are sovereign states. This result is mechanically identical to the predator-prey result (Domain E2, $b_{\text{mean}} = +0.102$): in both cases, complete node extinction would destroy the hub. The equivalence between political sovereignty and ecological mutual interdependence as braking mechanisms — across different substrates and temporal scales — was not predicted a priori and constitutes the most theoretically significant finding of the extended corpus.

N-body matrix: Mexico

Verification with INEGI 2022 data for the 32 Mexican federal entities confirms the system follows preferential attachment: $f(\text{rank}) = 396.8 \times \text{rank}^{(-0.473)}$, $R^2 = 0.838$, $p < 0.001$ (Fig 3A). The five-level taxonomy classifies: Level 0 (Mexico City, 14.8% of national GDP), Level 1 (9 secondary attractors, 41.0%), Level 2 (8 logistic bypass nodes, 20.2%), Level 3 (11 shadow nodes, 16.8%), and Exogenous Anomalies (3 entities, 4.3%).

The composite gradient analysis reveals a critical error in the binary model (Fig 3C). The binary model measured $w_{ij}(\text{Tlaxcala} \rightarrow \text{Puebla}) = 26.2\text{k MXN}$. The N-body matrix reveals $w_{ij}(\text{Tlaxcala} \rightarrow \text{Mexico City, long-range}) = 216.8\text{k MXN}$. Total composite gradient: 243.0k MXN. The binary model underestimated Tlaxcala's satellization by a factor of $9.3\times$. The 89.2% of extraction flows directly to Mexico City, bypassing Puebla. Any policy strategy targeting only the Tlaxcala–Puebla relationship addresses 10.8% of the actual problem. The first documented cases of successful leapfrog within the Mexican national system are Querétaro ($b = -0.155$, $R^2 = 0.782$, $p < 0.01$, via aerospace manufacturing) and Nuevo León ($b = -0.058$, $R^2 = 0.935$, $p < 0.001$, via export manufacturing).

Fig 3. N-body matrix of the Mexican national system (INEGI 2022, 32 federal entities). (A) Log-log scatter of GDP per capita against rank. Power-law fit: $f(r) = 396.8 \times r^{(-0.473)}$, $R^2 = 0.838$, $p < 0.001$. (B) GDP per capita distribution by SNT five-level taxonomy. (C) Tlaxcala composite

gradient: the binary model underestimated total satellization by 9.3×; 89.2% of the gradient flows directly to Mexico City.

Atomic Sovereignty Index

The ASI quantifies the degree of cognitive sovereignty of an Atomic Node from observable behavioral data. Formula: $ASI = (\delta H \times \alpha) / F$, where δH is the Shannon entropy of event-type distribution (normalized 0–1, measures diversity of problems solved), α is the proportion of autonomous events over total classified events (measures initiative), and F is the tanh-normalized coefficient of variation of daily performance (measures internal chaos). The threshold $ASI \geq 1.0$ was calibrated empirically as the Elite cohort median.

Validated on the HackerEarth 2026 dataset ($N = 4,774$ users, 409,287 events): 13 users (0.27%) reach $ASI \geq 1.0$. Classification precision: 1.0000 (zero false positives). Spearman correlation $ASI-CSI_V3$: $rs = 0.178$, $p < 0.001$ — the ASI captures a dimension complementary to the production-volume index. Component values by cohort are shown in Fig 4B. The top predictor of Elite-cohort trajectory is AI agent delegation (SHAP score ≈ 0.5), operationalizing the cognitive leapfrog: the platform dimension where accumulated hub advantage does not apply directly is AI-native operation.

Fig 4. Atomic Sovereignty Index (ASI) operationalized on HackerEarth 2026 behavioral data ($N = 4,774$ users, 409,287 events). (A) ASI distribution by cohort (Elite 0.5%, Intermediate 6.4%, Basic 93.1%). Dashed line: $ASI = 1.0$ (sovereignty threshold). Precision = 1.0000, zero false positives. (B) Mean ASI component values (δH : diversity; α : autonomy; F : internal entropy) by cohort. (C) Satellization cycle phases with empirically calibrated ASI thresholds.

DISCUSSION

Universality of the power law

The power-law form $R(t) = a \cdot t^b$ with b ranging from -2.852 to $+7.086$ across 502 cases in 11 domains and 30 orders of magnitude of temporal scale supports the claim that satellization is a universal organizing principle of complex systems. The mechanism requires only three conditions: (1) two nodes of asymmetric accumulated mass; (2) critical proximity; (3) a resource transfer mechanism. When these conditions are met, the power-law dynamic emerges regardless of substrate — trade routes, gravitational fields, predation, antibiotic selection, AI adoption, or political decree.

This universality connects to established theory in specific ways. The preferential attachment mechanism [3] predicts the distribution of degrees in scale-free networks; the SNT formalizes the directional flow of resources within such networks and quantifies its rate. Lotka-Volterra dynamics [20,21] predict oscillatory predator-prey ratios; the SNT correctly identifies predator-prey as a mutual-interdependence system producing $b \approx 0$. The Free Energy Principle [5] provides a neurological substrate for why individual Atomic Nodes with chaotic environments (high F) have lower probability of executing the leapfrog. Complex adaptive systems theory [6] describes emergence from local rules; the SNT identifies satellization as one specific recurring emergent dynamic with a mathematically predictable trajectory.

Institutional friction as the key variable

The most theoretically important finding is that institutional friction — not substrate, not temporal scale, not the magnitude of mass asymmetry — is the primary predictor of b . This has a direct implication for intervention design: the most effective way to slow satellization for a peripheral node is not to increase its mass (which is slow and expensive) but to increase its indispensability to the hub — creating institutional friction that makes complete extraction self-defeating.

The astrophysical extension of the corpus provides the most unexpected confirmation of this principle. The accretion mode of a black hole — an entirely physical phenomenon with no institutional component — produces a pattern identical to institutional friction in human systems. RIAF mode limits accretion efficiency via radiative thermodynamics; political sovereignty limits satellization via military resistance and economic interdependence; predator-prey interdependence limits extinction via the predator's own food supply. These are three implementations of the same braking principle in different substrates, all producing $b \approx 0$.

The N-body correction

The most practically important finding for regional development policy is the $9.3\times$ underestimation of Tlaxcala's satellization by the binary model. Standard regional economics models the inequality between two proximate nodes as a bilateral relationship. The N-body correction reveals that the long-range gradient (Tlaxcala $\rightarrow$ Mexico City) is $8.3\times$ larger than the direct gradient (Tlaxcala $\rightarrow$ Puebla), and that 89.2% of the total extraction flows directly to the hub center rather than to the immediate neighbor. This systematic error in bilateral models likely extends to other regions embedded in multi-level hierarchies, and has direct implications for the design of regional development interventions.

Limitations

Several limitations require acknowledgment. Historical city and country data carry $\pm 20\%$ uncertainty for pre-1820 estimates. The trigger classification (abrupt/gradual/hybrid) involves theoretical judgment calls that different researchers might resolve differently for ambiguous cases. Biological and astronomical domains have small N per domain ($n = 4\text{--}20$), limiting within-domain inference. The HackerEarth dataset is proprietary; ASI operationalization requires replication on independent behavioral datasets before being considered robust. The Micro (individual) and Macro (superorganism) modules of the Triple-Resolution Model have partial operationalization and require independent empirical validation.

The 31.1% significance rate reflects the structural inclusion of mutual-interdependence domains where $b \approx 0$ is the correct prediction, not a failure of the model. However, the proxy data quality varies substantially across domains, and several astronomical cases use synthetic mass-time series from formation models rather than direct observational measurements at multiple epochs. These cases are clearly flagged in the replication package [7].

Future directions

Four research lines follow directly from these findings. First, ASI operationalization on independent behavioral datasets from other platforms. Second, extension of the N-body matrix to other national systems to test whether the 89% long-range extraction finding generalizes. Third, mathematical formalization of the cascade transmission principle — how macro-scale events propagate to meso and micro levels with domain-dependent attenuation. Fourth, direct observational data for astrophysical cases currently relying on formation models, to increase the precision of domain-level estimates.

CONCLUSIONS

The power-law satellization mechanism — $R(t) = a \cdot t^b$ — operates universally across 502 cases in 11 domains spanning 30 orders of magnitude of temporal scale. Abrupt triggers produce faster satellization than gradual ones (Mann-Whitney $p = 1.91 \times 10^{-5}$, stable across three corpus expansions). Institutional friction is the dominant predictor of satellization velocity across all substrates. Political sovereignty and ecological mutual interdependence are mechanically equivalent braking mechanisms. The binary model underestimated Tlaxcala's regional satellization by 9.3×. The Atomic Sovereignty Index operationalizes cognitive sovereignty from behavioral data with precision = 1.0.

The most important implication is not theoretical but practical: a model that correctly identifies the rate of satellization, its primary drivers, and its failure taxonomy is a model that enables intervention before the event horizon is crossed. For Tlaxcala, 89.2% of the problem lies in its relationship with Mexico City, not Puebla. For any Atomic Node in 2026, the cognitive leapfrog — AI agent orchestration rather than linear execution — is the first dimension in recent history where the accumulated advantage of dominant nodes does not directly apply to new entrants.

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AUTHOR CONTRIBUTIONS

Conceptualization: EZC. Methodology: EZC. Software: EZC. Validation: EZC. Formal analysis: EZC. Investigation: EZC. Data curation: EZC. Writing — original draft: EZC. Writing — review & editing: EZC. Visualization: EZC. Project administration: EZC.

(CRediT taxonomy: <https://credit.niso.org/>)

DATA AVAILABILITY

All code, processed data, and figures are available in the replication package at Zenodo (DOI: 10.5281/zenodo.19446521) and GitHub (<https://github.com/Inzainos/The-shadow-Node-Theory>). The HackerEarth 2026 raw dataset is proprietary and not redistributable; aggregate results are

available in the replication package (snt_asi_scores.csv, snt_corpus_final.csv). The Maddison Project Database 2023 is available at <https://www.rug.nl/ggdc/historicaldevelopment/maddison/> under CC BY 4.0. INEGI data are available at <https://www.inegi.org.mx/temas/pib/>.

COMPETING INTERESTS

The author declares no competing interests.

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